Lec 18 Practice (Laplace)

Ex 1 $X(s) = \dfrac{7s-6}{s^2-s-6}$ find $x(t)$.

$$X(s) = \frac{7s-6}{s^2-3s+2s-6}$$

$-3 \times 2 = -6$
$-3+2 = -1$

$$= \frac{7s-6}{s(s-3)+2(s-3)} = \frac{7s-6}{(s+2)(s-3)} = \frac{k_1}{s+2} + \frac{k_2}{s-3}$$

cover up method:

$$k_1 = \left.\frac{7s-6}{\boxed{\cancel{(s+2)}}\,(s-3)}\right|_{s=-2} = \frac{-14-6}{-2-3} = 4$$

covered-up

similarly

$$k_2 = \left.\frac{7s-6}{(s+2)\boxed{\cancel{(s-3)}}}\right|_{s=3} = \frac{21-6}{3+2} = 3$$

$$X(s) = \frac{4}{s+2} + \frac{3}{s-3}$$

Quick check put $s=0$ (or any convenient value) to see original & expanded versions give same answer (note: though this does not guarantee correctness, it gives some confidence)

→ use Laplace table $\frac{1}{s+a} \Leftrightarrow e^{-at}u(t)$

$$\Rightarrow x(t) = \left(4e^{-2t} + 3e^{3t}\right)u(t)$$

Ex.2 $X(s) = \dfrac{2s^2+5}{s^2+3s+2}$, find $x(t)$.

→ This is $M=N$ case (same ~~highe~~ order in num and den)

→ shortcut for such cases:
separate out coefficient of highest power.

$$\boxed{\begin{array}{l} 2\times 1 = 2 \\ 2+1 = 3 \end{array}}$$

$$\Rightarrow X(s) = \frac{2s^2+5}{s^2+3s+2} = \frac{2s^2+5}{(s+1)(s+2)}$$

$$= 2 + \frac{K_1}{s+1} + \frac{K_2}{s+2}$$

→ now apply cover-up rules, to get.

$$K_1 = 7 \text{ and } K_2 = -13$$

$$\Rightarrow X(s) = 2 + \frac{7}{s+1} - \frac{13}{s+2}$$

→ Use Laplace table $\left\langle \begin{array}{l} \frac{1}{s+a} \Leftrightarrow e^{-at}u(t) \\ 1 \Leftrightarrow \delta(t) \end{array} \right.$

$$\Rightarrow x(t) = 2\delta(t) + \left(7e^{-t} - 13e^{-2t}\right)u(t)$$

Ex3 $X(s) = \dfrac{5e^{-2s}}{(s+1)(s+2)}$ find $x(t)$.

$\rightarrow$ we note that e^{-2s} represents delayed version.

e.g. if $x_1(t) \Longleftrightarrow X_1(s)$

then $x_1(t-2) \Longleftrightarrow e^{-2s} X_1(s)$

$$\Rightarrow X(s) = \underbrace{\left(\frac{5}{(s+1)(s+2)}\right)}_{X_1(s)}\left(e^{-2s}\right)$$

$$\Rightarrow X_1(s) = \frac{5}{(s+1)(s+2)} = \frac{5}{s+1} - \frac{5}{s+2} \quad \text{(PFE)}$$

$$\Rightarrow x_1(t) = \left(5e^{-t} - 5e^{-2t}\right)u(t)$$

$$\therefore x(t) = x_1(t-2) = \left(5e^{-(t-2)} - e^{-2(t-2)}\right)u(t-2)$$

Ex4 Given $e^{-at}\cos bt\, u(t) \Longleftrightarrow \dfrac{s+a}{(s+a)^2+b^2}$

Find $\mathcal{L}\{\cos bt\, u(t)\}$.

$\rightarrow$ set $a = 0$

$$\cos bt\, u(t) \Longleftrightarrow \frac{s}{s^2+b^2}$$

EX5 Assuming Zero ICs, ~~Solve DE~~ Find via $\mathcal{L}$-Tx.

$$\frac{d^2x(t)}{dt^2} = \delta(t) - 3\delta(t-2) + 2\delta(t-3)$$

$\Rightarrow$ Taking $\mathcal{L}$-Tx

$$s^2X(s) - s\cancel{x(0)}^{0} - \cancel{\dot{x}(0)}^{0} = 1 - 3e^{-2s} + 2e^{-3s}$$

$$\Rightarrow \quad X(s) = \frac{1}{s^2}\left(1 - 3e^{-2s} + 2e^{-3s}\right)$$

EX6 use $\mathcal{L}$-Tx to find the conv.

$$c(t) = \underbrace{e^{at}u(t)}_{x_1(t)} * \underbrace{e^{bt}u(t)}_{x_2(t)}$$

$$\rightarrow \quad x_1(t) \Longleftrightarrow \frac{1}{s-a}$$

$$x_2(t) \Longleftrightarrow \frac{1}{s-b}$$

Taking $\mathcal{L}$-Tx

$$\rightarrow \quad C(s) = X_1(s)\,X_2(s) = \frac{1}{(s-a)(s-b)}$$

using PFE $\rightarrow$

$$C(s) = \frac{1}{a-b}\left(\frac{1}{s-a} - \frac{1}{s-b}\right)$$

$$\Rightarrow c(t) = \frac{1}{a-b}\left(e^{at} - e^{bt}\right)u(t)$$